Can a Cheap Text Monitor Tell Right from Wrong Reasoning?

CSI 436/536 Machine Learning (Fall 2026) — Project P Midterm One-Pager Group: Jiayu Liao, Zeyu Pan, Yuhan Sun Oct 20, 2026

Research question (fixed by the project).

Do pretrained representations improve downstream ML tasks, particularly under limited training data? We turn the yes/no question into a measurement problem. With the class algorithm g and training data fixed, the *only* varied factor is the feature source φ, and we report the gap function

$$\Delta(\theta) \;=\; L_{\text{test}}\big(g \circ \varphi_{\text{simple}};\, \theta\big) \;-\; L_{\text{test}}\big(g \circ \varphi_{\text{pretrained}};\, \theta\big), \qquad \theta = (\text{train size } n,\ \text{distribution shift},\ \text{perturbation}).$$

Task and data.

A cheap *text monitor* for math reasoning: given a problem and one solution step, classify the step as correct or incorrect. **PRM800K** (OpenAI, MIT): human-labeled step-level correctness on MATH solutions — 911,867 labeled steps (27.9% incorrect) after our data preparation; grouped by problem into 632,422 train / 190,573 test steps. **Math-Shepherd** (GSM8K, automatically labeled, 613,666 steps): transfer test set only. Rule-based perturbed variants (typos, number formatting, LaTeX style): robustness test sets. Labels come with the datasets; no LLM labeling is used.

Method (strictly within the rules).

Pretrained embeddings — `all-MiniLM-L6-v2`, `bge-base`, `bge-large` — are *frozen* feature extractors only (forward pass, no fine-tuning). On top we train only models covered in class: logistic regression (L7) and linear SVM (L11). Control arm: TF-IDF (word / char n-grams) with the same classifiers. Protocol: input = problem + current step; **group split by problem** (no step of one problem crosses train/test); primary metrics **balanced accuracy / AUROC** (labels are imbalanced); 10 stratified re-samples per n, mean ± CI; C selected by grouped k-fold CV inside the training set; test set touched once; all seeds fixed.

Current progress (all three analysis axes complete).

$\Delta(n)$**, sample efficiency**: at n=20 all features sit near chance (≈ 0.51 AUROC); from n=50 TF-IDF pulls ahead and the gap grows monotonically; at full data $\Delta = -0.045$ to -0.064 (SVM, CI excludes 0). *In-domain, pretrained representations do not improve this task at any n* — a boundary we report as-is. Δ(**shift**)**, generalization**: training on MATH → testing on GSM8K flips the sign: bge-large $\Delta = +0.020$, CI $[0.018, 0.021]$; bge-base $+0.006$, CI $[0.004, 0.008]$ (paired bootstrap; MiniLM n.s.). Embeddings also overfit far less: train–test AUROC gap 0.040–0.065 vs. TF-IDF's 0.153–0.168. Δ(**noise**)**, robustness**: under rule-based perturbations all degradations are small (≤ 0.007 AUROC), and embeddings degrade no more than TF-IDF. Headline: *the value of pretrained features is regime-dependent — absent in-distribution, present under shift.* Robustness checks: a 50/50 re-sampled training set and `class_weight=balanced` change AUROC by < 0.01.

Challenges and plans.

Early on we caught a label-construction bug of our own (taking only the `chosen` completion per step yielded 99.8% positives and degenerate metrics); we rebuilt the data from all human-rated completions and re-ran everything — all numbers above are the corrected version. Remaining care: the MATH→GSM8K shift is a *compound* shift (problem type, source, labeling method), so we report it without single-factor attribution. Plans: final report written as a measurement report (summary effect sizes + three boundary curves); error analysis on the hardest step types; an input ablation (step only vs. problem+step).

AI use disclosure (per syllabus §4.8).

AI assistants were used only for language polishing and code debugging. All experimental design, data processing, results, analysis, and conclusions were produced and independently verified by the group.